

3 (**long**) Figure 3 shows the entrance part of the water flow in a horizontal circular pipe of constant radius  $R$ . The pipe wall boundary layer develops immediately downstream of the entrance (plane 0), where the flow is uniform with velocity  $V_0$ . It reaches the centre of the pipe at plane 1 so that an inviscid core no longer exists downstream of this point. At plane 1, the flow has an axisymmetric parabolic velocity profile,

$$V(r) = V_c \left[ 1 - \left( \frac{r}{R} \right)^2 \right]$$

where  $V$  is the velocity at radius  $r$  and  $V_c$  is the velocity at the centre line of the pipe. The density of water is  $\rho$ .

- (a) Find  $V_c$  in terms of  $V_0$ . [6]
- (b) Explain why Bernoulli's equation can be used to determine the pressure difference  $(p_0 - p_1)$  between plane 0 and plane 1. Use Bernoulli's equation to find  $(p_0 - p_1)$  in terms of  $\rho$  and  $V_0$ . [8]
- (c) Find expressions for the momentum fluxes at plane 0 and plane 1 in terms of  $R$ ,  $V_0$  and  $\rho$ . [10]
- (d) Find an expression for the horizontal force that the flow exerts on the pipe between plane 0 and plane 1. Comment on the physical origin of this force. [6]

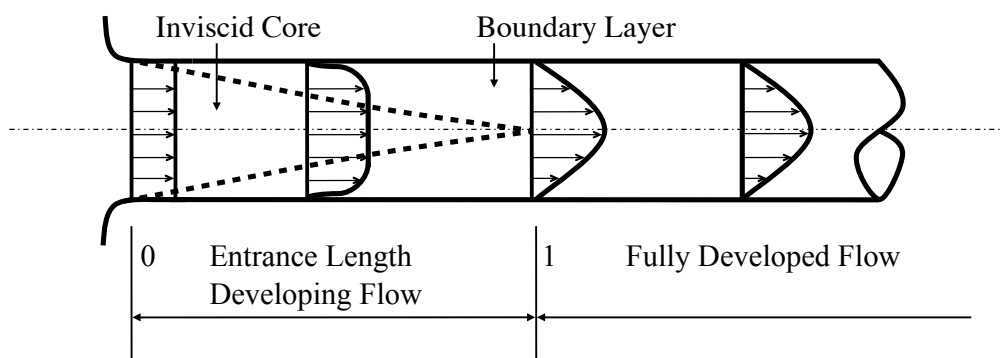


Fig. 3